

ELECTROCHEMISTRY

Machine learning–driven discovery of optimal designs for water electrolysis devices

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Green hydrogen production via water electrolysis is key to decarbonizing energy systems, but current electrolyzer designs are energy-intensive due to gas bubble entrapment and manual optimization. This study introduces a machine learning (ML) strategy that autonomously designs high-efficiency electrolyzer flow channels. We identify an array-type channel geometry that enhances bubble removal using a mixture-of-experts framework, with a parametric analysis establishing structure-performance relationships. A prototype electrolyzer incorporating the artificial intelligence–optimized channel demonstrated ~23% improvement in current density at 2 V compared to a conventional serpentine design. This enhancement was consistently in scaled-up devices, underscoring the effectiveness of the design across scales. In contrast to computational fluid dynamics–based approaches simulating every geometry, our ML surrogate performs data-driven screening to efficiently identify and validate high-performance channel structures. By decoding the relationships between topological features and multiphase transport, this work outlines a scalable pathway toward autonomous design of next-generation electrochemical systems.

INTRODUCTION

As global energy demand continues to rise alongside escalating environmental concerns, the need for clean and sustainable energy solutions has become increasingly pressing (1). Among emerging energy technologies, hydrogen is a pivotal component of future energy systems due to its clean nature, high energy content, and versatile applications (2). Water electrolysis—particularly the alkaline variant—has emerged as the primary pathway for large-scale green hydrogen production owing to its technological maturity, cost effectiveness, and long-term stability (3, 4). However, the feasibility of green hydrogen production at scale is fundamentally constrained by the high energy requirements of water electrolysis. Therefore, reducing energy consumption in hydrogen production is crucial for the practical and widespread deployment of green hydrogen. A proven approach to advance the performance of water electrolysis involves developing electrocatalysts with superior catalytic activity and using high-performance membranes (5–7). These innovations can lower reaction overpotentials and accelerate reaction kinetics (5, 8), thereby reducing energy losses and enhancing the overall efficiency of the water electrolysis process. Moreover, beyond improvements to catalysts and membranes, optimizing gas-liquid mass transfer within the electrolyzer is equally critical for minimizing the overall energy consumption in hydrogen production (9).

Basically, enhancing the gas-liquid mass transfer efficiency in water electrolysis fundamentally relies on the optimization of electrolyzer flow channel architectures. This critical requirement stems from the inherent gas-liquid two-phase flow characteristics during electrolysis, where the rapid removal of generated gas bubbles is critical to maintain electrode activity and preserve electrolyte conductivity (10). Therefore, the development of novel flow channel configurations for water electrolyzers has emerged as a research priority (11–13). Nevertheless, the design of highly efficient flow channels is complicated because of the

intrinsic complexity of two-phase flow dynamics, the high cost of experimental validation, and the substantial computational resources and time required for comprehensive numerical simulations (14). Besides, computational fluid dynamics (CFD)–only optimization is computationally expensive yet yields marginal experimental gains (15, 16). The rapid evolution of artificial intelligence (AI), progressing from early rule-based expert systems to sophisticated deep learning and neural network frameworks (17), has revolutionized optimization paradigms in energy systems (18–20). Modern AI demonstrates unparalleled capabilities in processing large-scale datasets, identifying complex nonlinear patterns, and performing multidimensional parameter optimization, which are the capacities that are crucial for reducing experimental iterations and accelerating technological innovation (21, 22). Recent advances in AI-enabled optimization frameworks have demonstrated marked breakthroughs in flow field design for redox flow batteries (RFBs), particularly in achieving optimized electrolyte distribution and enhanced energy efficiency (23). However, the extension of such AI-driven methodologies to multiphase flow reaction systems, exemplified by water electrolysis for hydrogen production, presents formidable scientific challenges. Unlike the inherent single-phase flow characteristics of RFBs (24), water electrolysis involves complex multiphysics phenomena including gas-liquid mass transfer dynamics and bubble nucleation kinetics (25, 26). Current AI architectures exhibit notable limitations in addressing such multifaceted challenges, particularly in achieving generalizability across multiscale operations and cross-configurational device architectures.

Across AI-assisted flow-field design for electrochemical systems such as alkaline water electrolyzers, proton exchange membrane (PEM) electrolyzers, and RFBs, prior work falls into two categories. The first relies on parameter-tuning surrogates that adjust a limited set of variables but do not produce manufacturable topologies (27–31). Despite strong machine learning (ML) fit metrics, these studies typically lack experimental validation, which undermines their credibility. The second category uses topology-level ML to directly propose channel geometries (23), although its proposals are often of limited variety that lacks representativeness. Furthermore, across both categories, a lack of interpretability analysis in previous studies offers little actionable guidance for optimizing flow channel designs (table S1).

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In this work, we proposed an automated platform for high-efficiency electrolyzer adaptive channel design (AHEAD) (Fig. 1 and fig. S1), which synergistically integrates expert-guided design, multiphysics simulations, ML model prediction, interpretable feature analysis, stochastic structure generation, and experimental validation through three-dimensional (3D) printing and high-speed imaging to systematically optimize electrolyzer flow channels (32). Specifically, we used multiphysics simulation to obtain a representative flow-channel structure database. Subsequently, we trained the models for the hydrogen production channel (MHPCs) to predict the water electrolysis performance across various geometric configurations of electrolyzer flow channels. Then, we constructed representative feature extraction and model interpretation pipeline (RFEMIP) to establish the correlations between flow channel characteristics and water electrolysis performance, thereby enabling the identification of critical design-driving features. This integrated approach facilitated the systematic screening of optimal flow channel configurations from a library of randomly generated electrolyzer flow channel designs. Last, by printing the AI-optimized flow channel

electrolyzer and conducting performance tests for water electrolysis, the results showed that the optimized array-type channel achieved an ~23% increase in current density at 2 V compared to the conventional serpentine design. The in situ high-speed camera further confirmed that the AI-screened array flow channels exhibit higher gas-liquid mass transfer efficiency. Furthermore, this enhancement was consistently observed in scaled-up devices, underscoring the design's effectiveness across different scales. This integrated, data-driven methodology not only accelerates the design and validation process but also underscores the potential of advanced computational and experimental techniques to enhance water electrolysis, which holds broad applicability across energy conversion and chemical processing industries.

RESULTS

Expert models for electrolyzer flow channels

Guided by expert knowledge, we used multiphysical simulations to design 1000 flow channel structures and calculated their current densities

Automated platform for high-efficiency electrolyzer adaptive channel design

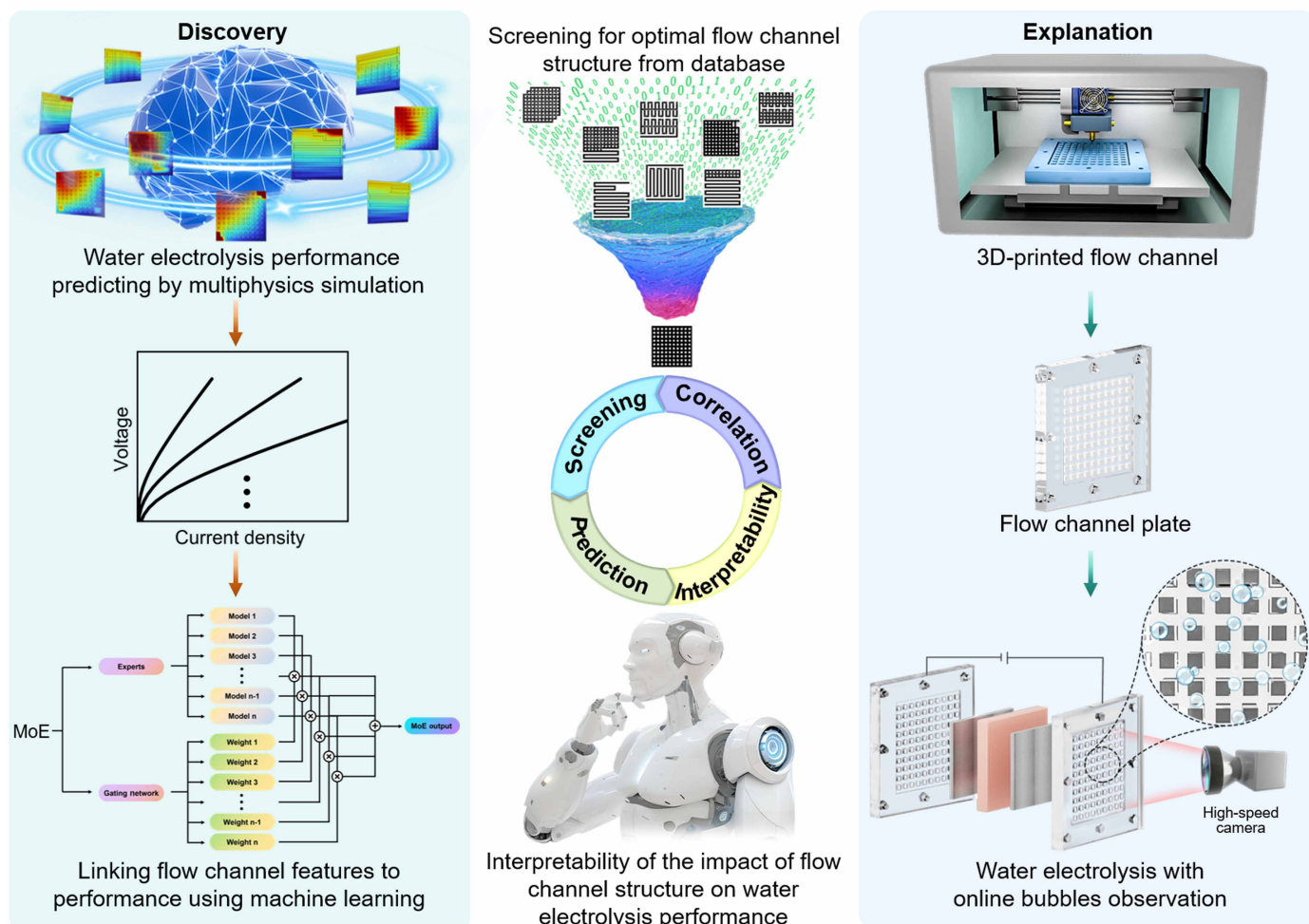


Fig. 1. The architecture of AHEAD. Our multiphysics simulations and ML integration revealed critical correlations between electrolyzer flow geometries and performance metrics. This enabled rapid identification of optimal channel architectures from computational designs, with additive manufacturing and high-speed imaging confirming the superior bubble management and efficiency of selected configurations. The illustrations were created using Cinema 4D, Adobe Illustrator, 3ds Max, and Microsoft PowerPoint.

under a voltage of 2.6 V, forming the basis of our initial dataset (the parameters are shown in table S2). The designed flow channel structures include single-channel and multichannel, with a distribution that provides a representative sampling of the design space (fig. S2), thereby establishing the foundation for the modeling process. On this basis, the MHPC framework was trained (table S3) to predict current densities across different flow channel configurations. The framework integrates multiple models, including the residual network (ResNet) (fig. S3) (33), vision transformer (ViT) (fig. S4) (34), swin transformer (Swin-T) (fig. S5) (35), graph isomorphism network (GIN) (fig. S6) (36), and mixture of experts (MoE) (fig. S7) (37). Specifically, for the ResNet, ViT, and Swin-T models, we adopted a pretraining plus fine-tuning strategy that delivers substantially better performance than training from scratch. The MoE model further builds upon this by ensembling the three best-performing pretrained models, namely, ResNet (res-18), ViT (base-16), and Swin-T (small) (fig. S8 and table S4), as its expert components.

To evaluate model performance, we first conducted experiments with varying training dataset sizes, ranging from 200 to 1000 samples. As the dataset size increased, model accuracy improved substantially (figs. S9 to S14). Notably, the performance metrics began to converge around 800 samples, suggesting that this dataset size is adequate to train a stable and reliable model. Based on this observation, 800 samples were used to train the models using fivefold cross-validation, while the remaining 200 previously unseen samples were held out as in-distribution (ID) data to assess the models' performance (figs. S15 to S20). In addition, an independent set of 50 out-of-distribution (OOD) samples was designed to further evaluate the model's generalization capability.

Among the models tested (Fig. 2, A to D, and table S5), Swin-T achieved the better performance, reaching an R^2 of 0.9869 and a mean absolute error (MAE) of 0.0200. These results underscored the effectiveness of Swin-T's hierarchical attention mechanism in capturing both local and global dependencies in intricate image data. ViT and MoE also performed well, with MAEs of 0.0228 and 0.0239, respectively, indicating strong capabilities in capturing complex patterns. ResNet also achieved solid performance (MAE = 0.0315). In contrast, GIN lagged behind (MAE = 0.0781), highlighting its limitations for pixel-level regression. All four models—ViT, Swin-T, ResNet, and MoE—demonstrate strong predictive accuracy, as reflected by their low mean absolute percentage error (MAPE) on both train and test sets (Fig. 2, E to H). Specifically, the average MAPE on the test set is 1.00% for ViT, 0.87% for Swin-T, 1.34% for ResNet, and 1.04% for MoE.

To evaluate the generalization capability of the models, we tested the well-trained models on both ID and OOD data (figs. S21 and S22), comparing predictions against CFD-calculated results. The analysis showed that Swin-T achieved the better performance on the ID data with an MAPE of 0.71% and MoE followed with an MAPE of 0.80% (Fig. 2, I to L, and fig. S23). The narrow distribution of residuals (Fig. 2M) reflects the model's stable predictive behavior during the training process. However, when tested on OOD data (Fig. 2N and fig. S24), the MoE model demonstrated the improved generalization capability, achieving an MAPE of 3.62%. It was followed by ResNet (4.14%), Swin-T (4.29%), and ViT (4.32%), all of which recorded MAPEs above 4%. These results prove the robustness of the MoE architecture across diverse structural patterns. It combines a set of heterogeneous expert models, while the gating network dynamically computes blending weights for each expert, enabling

adaptive fusion of diverse model capabilities. Consequently, extending the approach to PEM or other electrolyzer types requires only assembling system-specific training data and retraining the model. Moreover, our open-sourced MoE model for channel structure processing allows the community to customize expert modules according to specific task requirements, thereby effectively addressing specialized challenges in the field.

To extensively explore the design space of flow channel structures, we first proposed the local direction-limited random walk (LDL-RW) algorithm, a stochastic process modeled as a discrete-time Markov chain (38) on a discrete grid. Then, we used the LDL-RW algorithm to construct a large-scale library of candidate flow channel structures comprising 1 million design samples. This extensive candidate library offers an extensive sampling of the design space, providing robust data support for the subsequent selection of high-performance structures. Considered the superior generalization ability of the MoE model, we used it for the high-throughput calculation of current density across all candidate flow channel structures. The total of 1,000,000 predictions was then ranked in descending order to screen out the flow channel structures with the higher current densities. We then selected the top-ranked structures as promising design candidates and conducted further morphological and topological analyses, which revealed that these highest-current density structures exhibit an array-like channel distribution pattern. Through this comprehensive approach of large-scale generation and high-throughput prediction, we constructed the water electrolysis channel current density database (WECID) containing 10^6 candidate structures with the LDL-RW algorithm. As the largest international dataset, it provides a robust foundation for exploring the relationship between channel morphology and electrochemical performance. Analysis of these designs revealed that array-based configurations offer superior performance, rendering them optimal for high-efficiency applications.

Model interpretability and insights

In electrolyzer design, while high predictive accuracy is undoubtedly important, it remains insufficient without corresponding insight into how channel architecture governs current density distribution. Distinct configurations, such as array-type versus serpentine layouts, substantially alter local flow velocity, turbulence, and reactive interface area, all of which directly govern current density distribution. While traditional black-box models may achieve performance predictions, they fail to elucidate the underlying physical mechanisms driving these effects. To address this limitation, we proposed RFEMIP, combining deep learning-based visual analysis, ML-based feature importance analysis, and symbolic regression to uncover the governing mechanisms through which flow channel configuration affects electrolyzer performance. The deep learning module extracts complex nonlinear features from the data and uses visualization techniques to interpret the influence of flow channel geometry. The ML component establishes predictive models through feature engineering to qualitatively elucidate how various structural features impact current density. Last, symbolic regression derives mathematical expressions from these relationships, uncovering how key geometric parameters dictate electrochemical performance quantitatively.

The framework starts by processing raw electrolyzer channel structure data using a convolutional neural network (CNN) (39). In the initial convolutional layers, the CNN captures low-level geometric details, including shape edges, groove lines, and local textures,

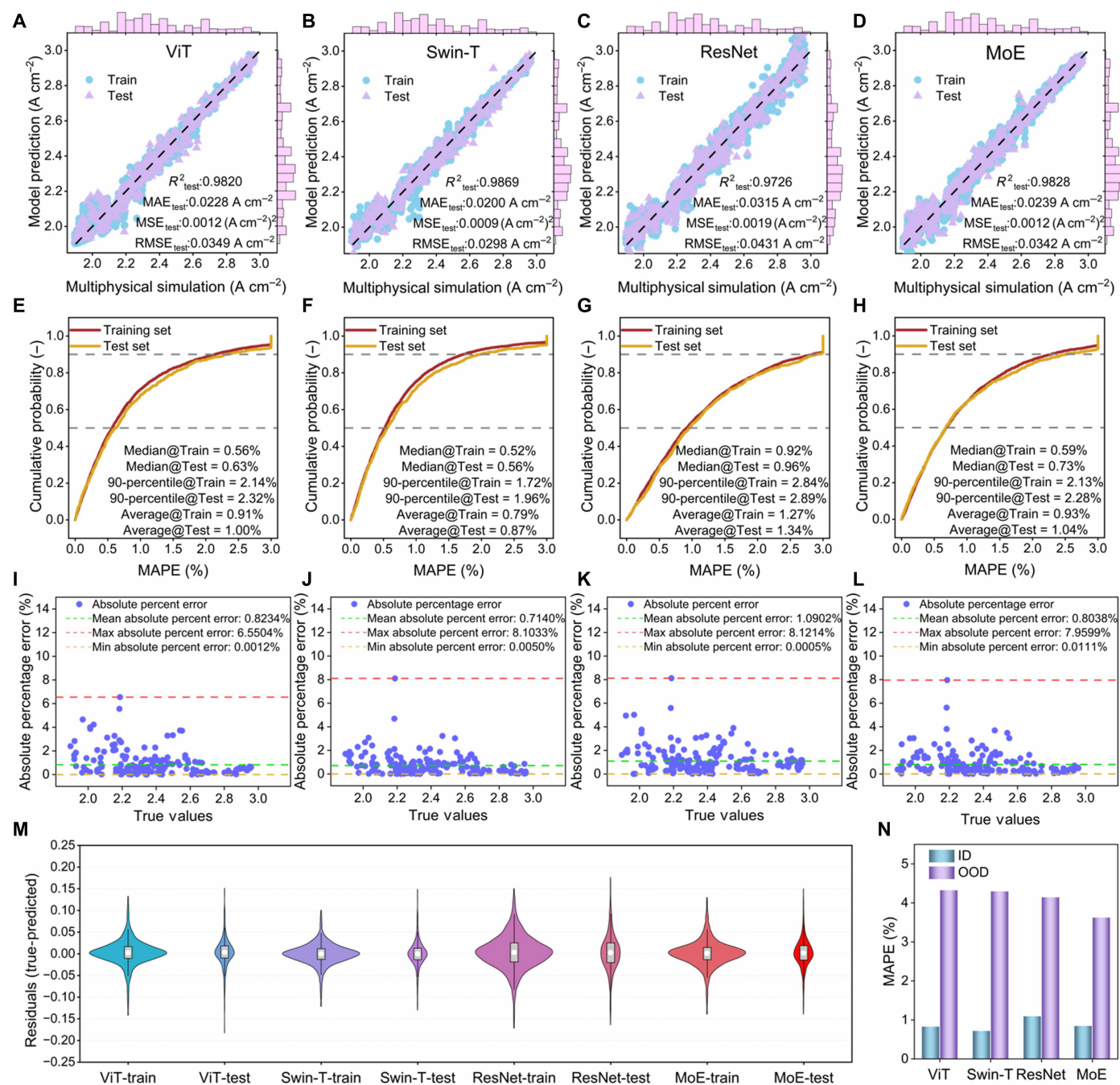


Fig. 2. Performance comparison of multiple deep learning models in multiphysical simulation for electrolyzer flow field design, based on the average outcomes of fivefold cross-validation; models were implemented using the following pretrained architectures: ViT (base-16), Swin-T (small), ResNet (res-18). (A to D) Predicted versus true multiphysical simulation outcomes using ViT (A), Swin-T (B), ResNet (C), and MoE (D) models. **(E to H)** Cumulative distribution of MAPE for ViT (E), Swin-T (F), ResNet (G), and MoE (H) models. **(I to L)** Scatter plots of absolute percentage errors on an independent ID test set for ViT (I), Swin-T (J), ResNet (K), and MoE (L) models. **(M)** Violin plots of residuals (true - predicted) for training and test sets across different models. **(N)** Test MAPE of different models on ID and OOD data.

which are later abstracted into more semantic, block-like patterns as the network deepens (figs. S25 to S27). To further understand these abstracted features, we applied gradient-weighted class activation mapping (Grad-CAM) (40) to visualize the regions most critical for predicting current density. Serpentine channels concentrate high-importance regions at bends and transition zones, reflecting localized

increases in fluid velocity, turbulence, and enhanced mass transfer. In contrast, array-based channels with their uniformly distributed units show evenly dispersed activation hotspots, indicating stable and balanced mass transfer conditions that contribute to higher current densities. To further underscore the contrast between the serpentine configuration and the array arrangement, we designed a

transitional Se-Ar channel that merges both structural patterns in one flow field, emphasizing the dominant role of uniform channel distribution in achieving higher current densities (Fig. 3A).

Building on these deep learning insights, we subsequently used ML-based feature engineering to extract and analyze quantifiable structural parameters. Single-channel flow fields are characterized by three main features. Area square (AS) represents the reactive interface area as the total number of grid cells occupied by the channel. Directional deviation (DD) reflects channel tortuosity as the average angular deviation of the flow path from a straight line. Number of corners (NC) is the count of direction changes in the channel path. In contrast, multichannel flow fields are described by AS and DD along with average path length (AL), number of spaces (NS), and number of branches (NB), which provide a comprehensive assessment of the flow field's configuration (table S6).

Correlation analysis identifies AS as the predominant factor driving current density (CD), since a larger AS boosts performance

by providing a greater electroactive surface area for the electrolysis reaction (Fig. 3B). To further investigate the relationship between structural features and current density, we constructed a series of ML models (figs. S28 to S40 and table S7). The performance evaluation—based on R^2 and root mean square error (RMSE)—demonstrated that BaggingRegressor ($R^2 = 0.9485$, $RMSE = 0.0404$ A cm $^{-2}$) is optimal for single-channel predictions (Fig. 3, C and D), while CatBoost ($R^2 = 0.9682$, $RMSE = 0.0451$ A cm $^{-2}$) excels in the multichannel context (Fig. 3, E and F). The optimal models are further validated by a comprehensive assessment of their overall performance in the Taylor diagrams (Fig. 3G). Based on the best-performing models, SHapley Additive exPlanations (SHAP) (41–43) analysis reveals that in single-channel configurations (Fig. 3, H and I, and fig. S41), increasing AS, NC, and DD shifts their SHAP values from near zero or negative to positive, indicating that a moderate increase in these geometric features can enhance current density. However, when assessing their relative importance, DD has a comparatively

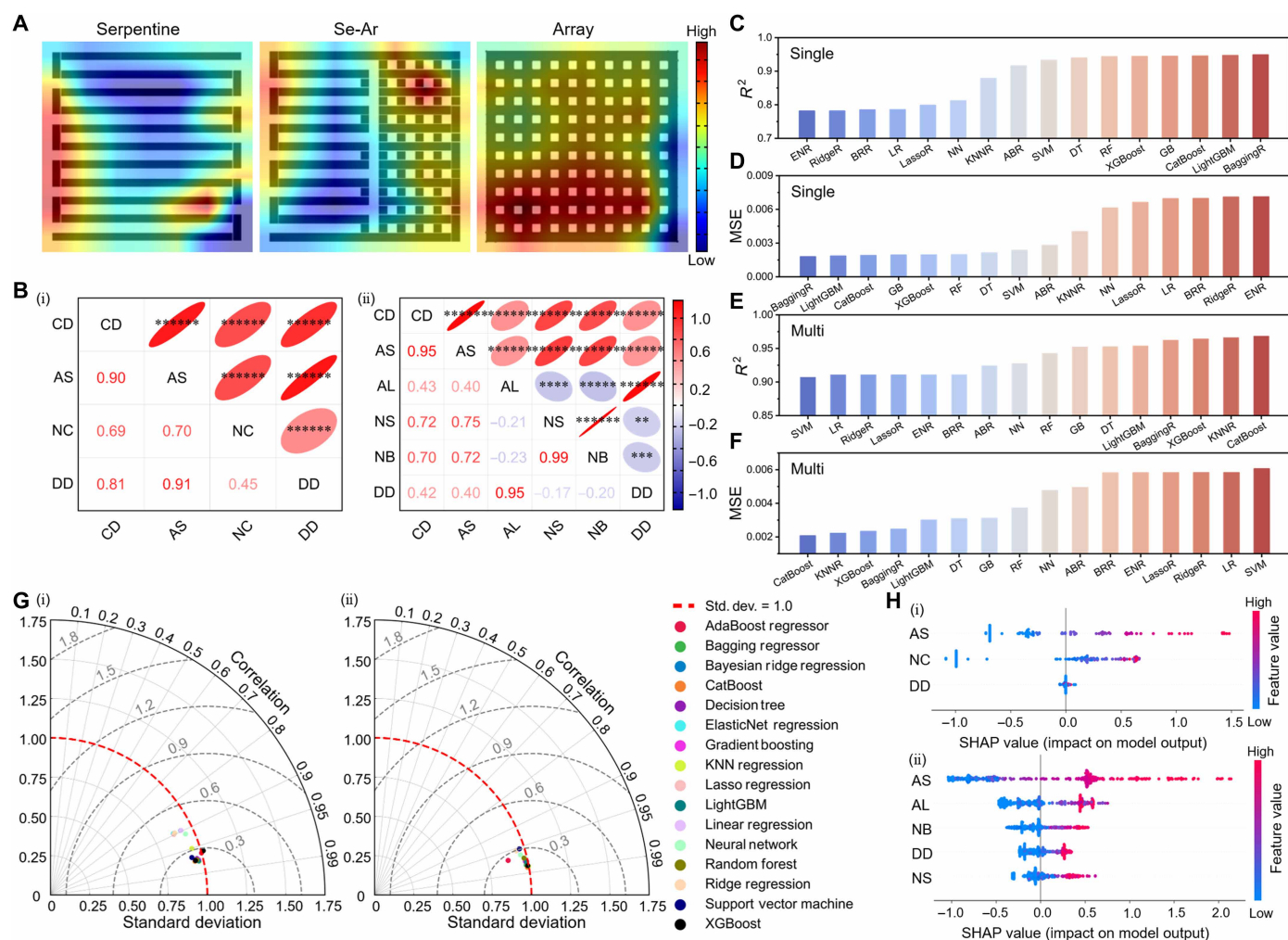


Fig. 3. Interpretability analysis of ML models for current density prediction in water electrolyzer flow field design. (A) Grad-CAM heatmaps illustrating the influence of local structural features on current density for flow channel configurations: (i) serpentine, (ii) Se-Ar, and (iii) array designs. (B) Correlation matrices for feature labels in (i) single-channel and (ii) multichannel structures, showing relationships between features. CD, current density. $*P \leq 0.01$; $**P \leq 0.001$; $***P \leq 0.0001$; $****P \leq 0.00001$; $*****P \leq 0.000001$; $*****P \leq 0.0000001$, Pearson correlation. (C to F) Comparative bar charts of R^2 and MSE for 16 ML models in single-channel [(C) and (D)] and multichannel [(E) and (F)] prediction tasks. (G) Taylor diagrams summarizing model performance metrics for (i) single-channel and (ii) multichannel tasks. (H) SHAP value analyses highlighting feature contributions (AS, NC, DD, etc.) to predicted current density in single-channel (i) and multichannel (ii) contexts.

smaller impact. In multichannel designs (Fig. 3H-ii and fig. S42), AS remains the most influential factor, while AL, NS, NB, and DD also contribute positively. Meanwhile, DD continues to exhibit a minimal effect on current density. The relatively minor impact of DD on current density may be attributed to excessive deviation increasing flow resistance and pressure drop, which offsets potential mass transport benefits.

Last, to quantitatively investigate these findings into a mathematical description, we used the sure independence screening and sparsifying operator (SISSO) (44) method for symbolic regression (tables S8 to S15). The regression results show that the relation between current density and its features of single-channel was as follows within the explored parameter space

$$CD = 2.33 - \frac{NC}{[1 + (NC)^2]} + 0.000787e^{(AS)^{1/3}} \quad (1)$$

The regression results indicate that AS has a strong exponential positive effect on current density, meaning that within a certain range, increasing AS can substantially enhance current density. Furthermore, an increase in the NC is associated with a rise in current density, confirming a positive correlation between CD and NC, which is consistent with the correlation observed in our SHAP analysis. The positive effect of NC on current density decreases with its rising value. Thus, the saturating effect of NC suggests that additional corners enhance turbulence and mixing for better bubble detachment, but excessive complexity may increase flow resistance and pressure drops, offsetting benefits.

For multichannel, the relation between current density and its features was also concluded

$$CD = 1.99 + 0.000947(2AS + AL) + 0.834 \frac{NB}{AS - AL} \quad (2)$$

Regression results show that in multichannel flow structures, AS and AL both positively influence current density, with AS exerting a stronger effect. In a multichannel design, the value of AS is always greater than AL, ensuring that the term $(AS - AL)$ is always positive, so an increasing NB can further boost current density. The contribution of NB highlights that larger branching structures facilitate more uniform flow distribution and mitigate stagnant zones, underscoring the importance of flow uniformity in minimizing bubble accumulation. Consequently, when optimizing multichannel designs, it is essential to balance AS, AL, and NB to achieve an optimal trade-off.

RFEMIP shows that AS is the dominant determinant of current density across single- and multichannel flow fields. In both systems, Grad-CAM localizes maximal activation at channel corners, consistent with a mechanism in which adding NC intensifies turbulence and mass transfer to raise current density. These effects, along with the influence of AL and NB, are consistently validated by both SHAP analysis and SISSO symbolic regression. Furthermore, it is worth noting that, whether in single-channel or multichannel analyses, DD does not appear in the symbolic regression expressions, indicating a relatively minor influence in both configurations. This finding is consistent with our SHAP analysis, offering mutual validation for the reliability of both methods. This integrated pipeline comprising CNN-based feature visualization, interpretable modeling, and SISSO symbolic regression forms a cohesive framework that explains how flow channel structure affects current density, providing data-driven insights for optimizing flow-field designs within the studied parameter space.

Experimental validation

To validate the ML and numerical simulation results (figs. S43 to S46), we used metal 3D printing (figs. S47 and S48) to fabricate three representative electrolyzer flow-channel designs (Fig. 4A). The water electrolysis performance of each design was then evaluated using a commercial membrane and catalysts (Fig. 4B). The current-voltage relation (I - V) curves measured at 50° to 80°C (Fig. 4, C to E, and fig. S49) show that at any given voltage, the electrolyzer with the array flow channels achieves a notably higher current density than those with serpentine or Se-Ar channels. For example, at 80°C and 2.0 V, the serpentine design delivers 0.7846 A cm⁻², whereas the array design reaches 0.9603 A cm⁻². This performance gap (current density differences) becomes even more pronounced at higher voltages and temperatures (fig. S50).

Figure 4F compares the current densities of the three designs at 2.0 V over a range of temperatures, highlighting the improvements achieved through flow-channel optimization. As temperature increases, the advantage of the array design grows: At 2.0 V and 80°C, the array design delivers a 22.4% higher current density than the traditional serpentine design. Figure 4G shows the cell voltage required at 80°C to achieve various current densities, emphasizing the reduced voltage needed with the optimized flow channels. For instance, at 0.3 A cm⁻² and 80°C, the array design requires a 5.5% lower cell voltage than the serpentine design, indicating enhanced efficiency. To further probe internal resistance differences, electrochemical impedance spectroscopy (EIS) was performed, and the results are presented as Nyquist plots (Fig. 4H). The array design exhibits a lower ohmic resistance, $R = 0.028 \Omega \text{ cm}^2$, compared to the serpentine ($0.032 \Omega \text{ cm}^2$) and Se-Ar ($0.030 \Omega \text{ cm}^2$) designs, indicating improved mass transport, conductivity, kinetics, and surface reactivity (fig. S51 and table S16). These EIS results confirm the superior transport characteristics of the array design, reinforcing its advantages over the traditional serpentine design.

Visual platform for demonstration

Generally, superior flow channel performance stems from efficient internal gas-liquid exchange. However, directly observing this process in conventional flow channels is challenging due to design limitations. To address this and investigate gas-liquid mass transfer in detail, we fabricated transparent flow channels and used a high-speed camera to in situ observe water electrolysis (Fig. 5A and figs. S52 and S53). This setup allowed real-time visualization of bubble behavior, tracking their generation, growth, and movement through the channels. Figure 5B shows snapshots of bubbles escaping over a 1-s interval (0 to 1 s) in three different flow channel configurations. Bubble size and residence time vary markedly across these designs, and we highlighted representative bubbles (green dashed outlines) in each case for analysis. In the serpentine channel, the highlighted bubble remains essentially unchanged in size and position from 0 to 1 s. In the array channel, bubbles initially accumulate in the lower-left corner at 0 s and then gradually disperse toward the upper right; by 1 s, most of these bubbles have been carried out of the channel. For a direct comparison between these extremes, we also designed a hybrid configuration (Se-Ar flow channel). In this intermediate design, the serpentine sections tend to produce larger bubbles while the array sections produce smaller ones. Over the 0- to 1-s interval, the smaller bubbles traverse the channel much more easily, whereas larger bubbles often become trapped and form blockages that impede

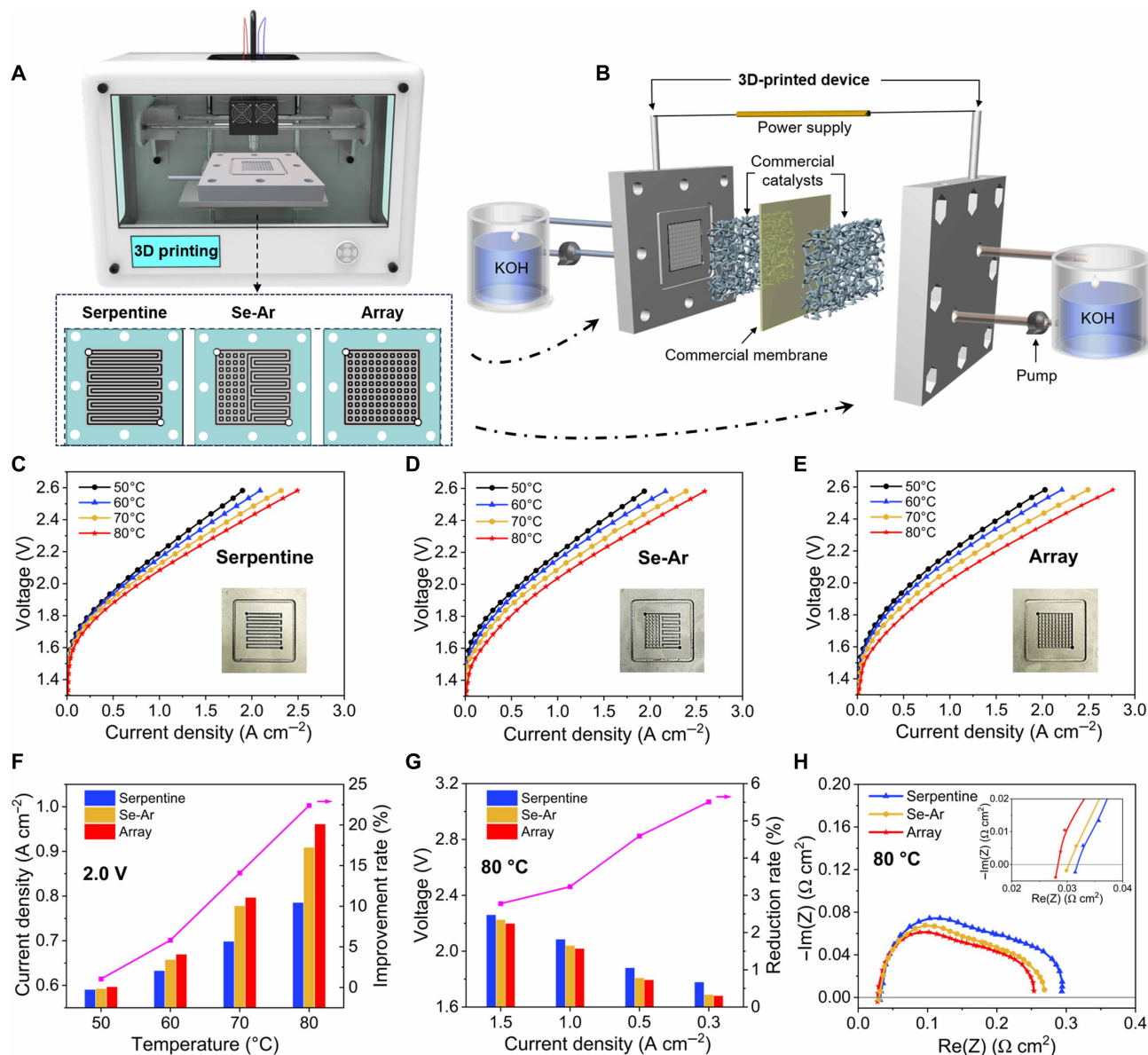


Fig. 4. Fabrication and electrochemical testing of three different water electrolyzer designs. (A) 3D-printed water electrolyzer devices with three different designs (serpentine, Se-Ar, and array flow field). The illustration was created using Cinema 4D and Microsoft PowerPoint. (B) Schematic diagram of alkaline water electrolysis assembled with the devices obtained through 3D printing, commercial catalysts, and commercial membrane. The illustration was created using Cinema 4D and Microsoft PowerPoint. (C) *I*-*V* curves of four different temperatures with 30 wt % KOH feed of serpentine flow field. (D) *I*-*V* curves of four different temperatures with 30 wt % KOH feed of Se-Ar flow field. (E) *I*-*V* curves of four different temperatures with 30 wt % KOH feed of array flow field. (F) The current density differences and improvement rate for three flow channels were examined at 2.0 V across a range of temperatures. (G) The voltage differences and relative reduction rate for three flow channels were examined at 80 °C across a range of current densities. (H) EIS spectra of three flow channels at 80 °C.

gas release. In effect, small bubbles can travel quickly from the inlet to the outlet, minimizing mass-transfer resistance in the channel.

Figure 5C shows the relative frequency distribution of bubble cross-sectional areas for the three flow channel designs. All three configurations are dominated by small bubbles (cross-sectional area in the range 0.1 to 0.3 cm²), but the array channel produces the highest proportion of these small bubbles (74.4% of the total), compared to 64.5% in the Se-Ar hybrid and 51.0% in the serpentine design. In contrast, comparatively large bubbles (>0.5 cm²) are least common in the array channel (11.6%), whereas they occur more

frequently in the serpentine (22.4%) and Se-Ar (12.9%) channels. The elevated occurrence of large bubbles in the Se-Ar channel is likely due to bubble accumulation in its serpentine sections; there, bubbles coalesce into larger ones that cannot be expelled within the short observation window, leading to a higher fraction of large bubbles in that configuration.

We also analyzed bubble residence times across the three flow channel configurations (Fig. 5D). Unlike the case for bubble cross-sectional area, the distribution of bubble velocities was distinctly bimodal, with two prominent peaks corresponding to slow (<10 cm

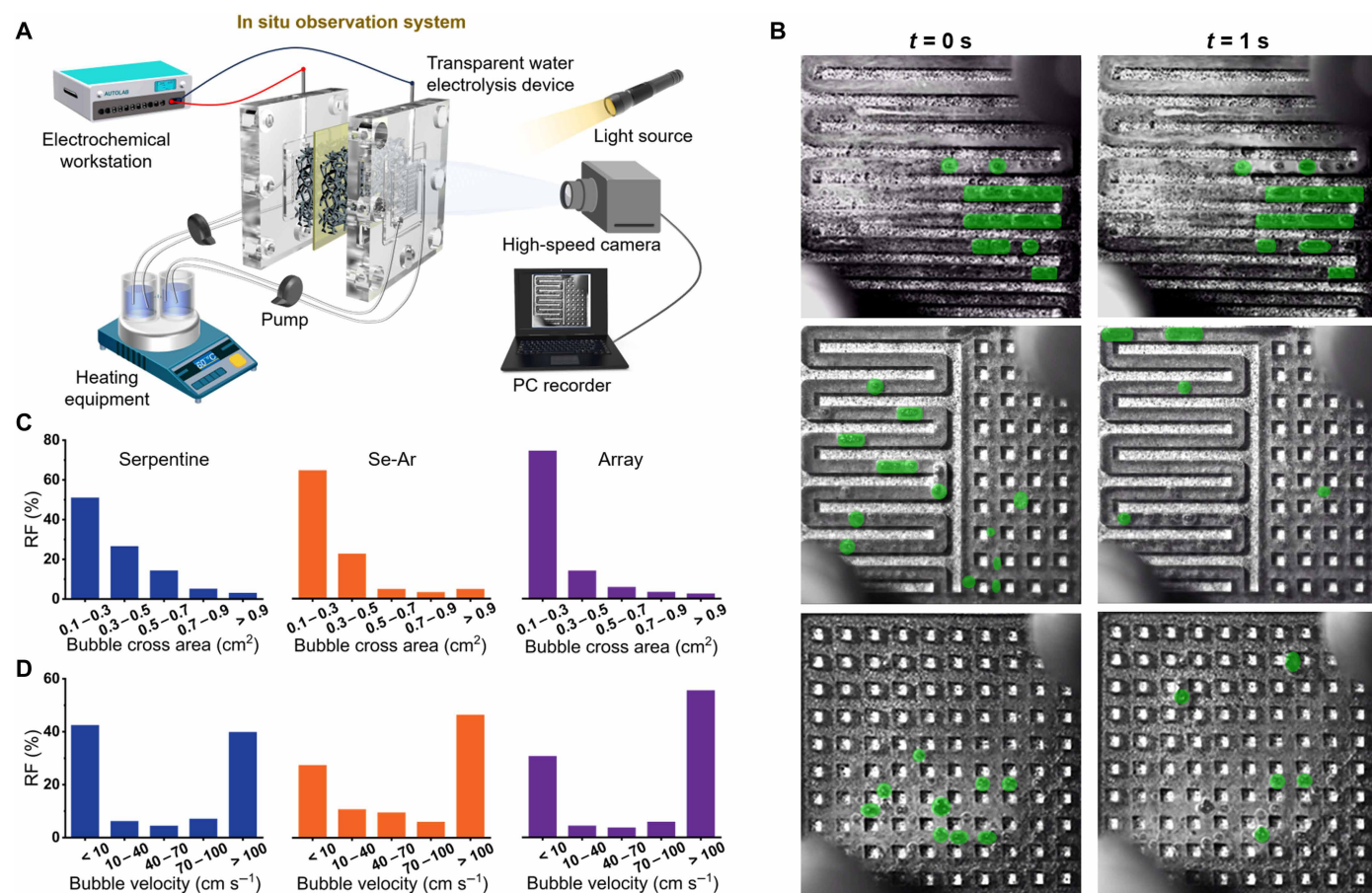


Fig. 5. In situ observation system for water electrolysis process. (A) Schematic in situ observation system for bubble generation during the water electrolysis process. The illustration was created using Cinema 4D and Microsoft PowerPoint. (B) The movement of bubbles in the three different flow channels was observed, tracking their position changes over time. (C) Relative frequency distribution of bubble cross areas in the three different flow channels. (D) Relative frequency distribution of bubble velocity in the three different flow channels.

s⁻¹) and fast (>100 cm s⁻¹) motion. Fast-moving bubbles traversed the channel without coalescing, allowing them to reach the outlet quickly, whereas slower bubbles tended to coalesce into larger ones that impeded their own movement. Quantitatively, the array configuration produced the largest fraction of fast-moving bubbles (55.5%, versus 30.7% slow-moving), compared to 39.8% fast (42.5% slow) in the serpentine channel and 46.4% fast (27.4% slow) in the hybrid Se-Ar channel. These observations are consistent with the overall faster bubble transport observed in the array design. Notably, larger bubbles and prolonged retention times increase the system's dead volume, which in turn reduces mass transfer efficiency and ultimately impairs water electrolysis performance. The dynamic processes of bubble generation and detachment are further illustrated in movies S1 to S3.

In the serpentine flow channel, gas-liquid mass transfer is hindered because bubbles generated at the inlet tend to coalesce with those formed further along the channel, causing them to grow larger and potentially block the flow path. This issue is exacerbated at higher current densities. By contrast, the array flow channel mitigates this problem by breaking larger bubbles into smaller ones as they pass through each block of the array. This bubble fragmentation improves dispersion and reduces retention time, thereby enhancing

mass transfer efficiency. The direct side-by-side comparison provided by the Se-Ar channel further highlights this contrast. On the serpentine side of this hybrid channel, bubbles coalesce and grow, whereas on the array side, they break up and disperse much more readily. This stark difference underscores the superior mass transfer performance of the array configuration.

Scaled-up experiments

To validate the applicability of the AI-optimized array architecture in a large-sized water electrolyzer, we conducted the experiments with the increased active area of 100 cm² (Fig. 6, A and B). Figure 6C shows the assembled water electrolysis system, where synchronous voltage monitoring was performed using precision voltammetry coupled with a high-current potentiostat to minimize measurement artifacts. The experimental results reveal that under the conditions of 80°C and 2.0 V, the array-configuration electrolyzer achieved an ~58.3% higher current density and demonstrated superior long-term stability compared to the serpentine-design counterpart (Fig. 6D and fig. S54). This enhancement is notably more pronounced than in previous small-sized electrolyzers, which we attribute to a scaling effect: As the flow channel area increases, the gas bubble accumulation in serpentine channels becomes more severe, leading to notable

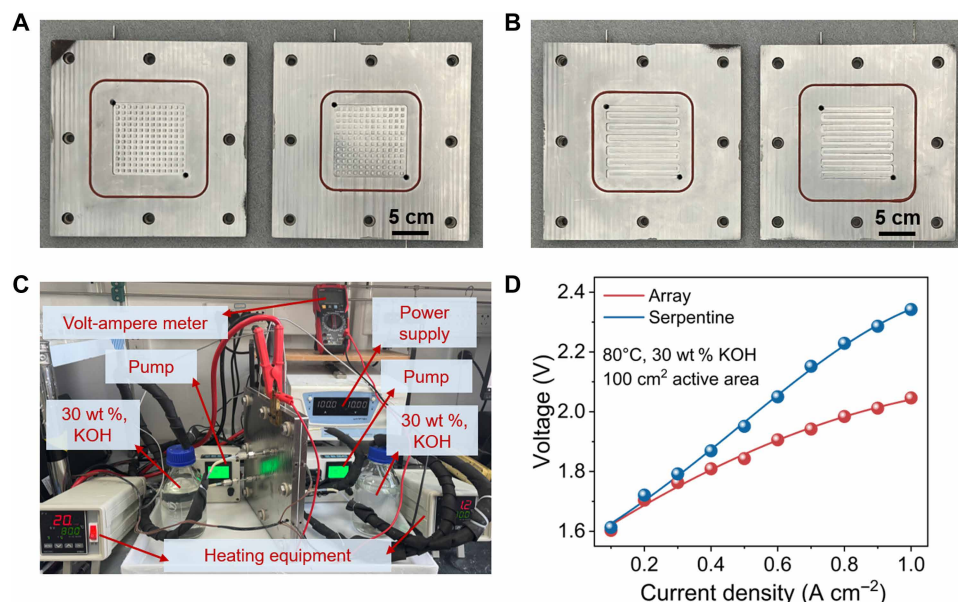


Fig. 6. Scaled-up experimental validation. (A and B) Photographs of array-type and serpentine flow channels with an effective area of $100\ cm^2$, respectively. (C) Photograph of the water electrolysis devices equipped with the large-sized flow channel. (D) I-V curves of the electrolyzers with large-sized array and serpentine flow channels ($80^\circ C$, 30 wt % KOH).

voltage spikes (45–47). In contrast, the array-configuration design maintains efficient bubble dispersion even at larger dimensions. These findings further demonstrate the superiority of AI-optimized flow channel designs. Simulation results also confirm that the array configuration outperforms the serpentine configuration in large-sized water electrolyzers (fig. S55).

DISCUSSION

We proposed AHEAD framework to construct an array flow channel that markedly improved water electrolysis performance. Theoretical calculations and numerical simulations predicted that this channel would provide substantially better gas-liquid mass transfer than conventional serpentine flow channels. To validate these predictions, we 3D-printed a prototype of the array flow channel designed by our ML model and tested it in a water electrolysis cell. At the same current density, the array channel required around 23% lower current density than a traditional serpentine flow channel. In situ visualization of bubble dynamics revealed that gas bubbles in the array channel detached and dispersed much faster, thereby alleviating gas blockage and further enhancing the overall electrolysis process, which was further confirmed by scaled-up experiments. This behavior is consistent with our model predictions and simulation results. Benchmark results confirm that the pipeline substantially accelerates design compared to laboratory testing (fig. S56). The AHEAD framework is transferable beyond alkaline electrolyzers, as it optimizes topology rather than a limited set of geometric parameters. Extending it to other systems, including PEM electrolyzers and RFBs, only requires system-specific data and MoE retraining. Overall, this study demonstrates that coupling ML with numerical simulations represents a powerful strategy for optimizing reactor configurations and guiding device design in chemical processes. Notably, this integrated approach not only enhances the efficiency of

individual processes but also contributes to advancing sustainable energy solutions, aligning with the global goal of decarbonizing energy systems and facilitating the transition to green hydrogen production.

MATERIALS AND METHODS

Materials

Potassium hydroxide (KOH) was obtained from Shanghai Meryer Biochemical Technology Co., Ltd. Commercial water electrolysis anode catalyst nickel-iron (NiFe) foam and cathode catalyst nickel-molybdenum (NiMo) foam were purchased from Xingzhenghong Technology Company. The porous membrane used in the experiment is a commercial membrane manufactured by Shenzhen KeXin Hydrogen Energy Co., Ltd. The water electrolyzer devices were prepared by using a 3D printing machine (HBD 350, Shanghai Hanbang United 3D Tech Co., Ltd.) The metal powder is stainless steel 316L.

Numerical simulation

For accurate modeling of water electrolysis, integrating electrochemical and fluid dynamics in multiphysics simulations is essential. The electrochemical process occurs within a water electrolyzer, where water dissociation is accompanied by phase changes. Consequently, fluid flow modeling must account for not only liquid electrolyte flow but also the coupled phase transfer phenomena. More specifics can be found in the Supplementary Materials.

Machine learning

Under expert guidance, we designed 1000 flow channel structures that provide a representative sampling of the design space and used multiphysics simulations to compute current densities at a specific voltage, forming the initial dataset. Building on this, we trained the MHPCs, including ResNet (33), ViT (34), Swin-T (35), GIN (36), and MoE (37), to predict current densities for various configurations.

We then developed the RFEMIP, which integrates deep learning, ML, and symbolic regression to reveal how key geometric parameters (such as channel area, NC, and DD) affect current density. In addition, the LDL-RW algorithm was used to generate 1 million diverse flow channel structures under design constraints, and the MoE model enabled high-throughput computation to establish the world's largest WECID. Our analysis indicates that array-based configurations offer superior high-efficiency performance. Specific modeling methods and training procedures can be found in the Supplementary Materials.

Alkaline water electrolysis test

The water electrolysis performance test was conducted according to the previous reports (48). The membrane used in the water electrolysis process was provided by Shenzhen KeXin Hydrogen Energy Co., Ltd. The electrolysis performance was obtained by 3D-printed water electrolyzer device with an effective membrane area of 4 cm². Commercial catalyst NiFe foam was used in the anode, while the cathode was NiMo foam. The electrolyte was 30 wt % KOH solution with a flow rate of 100 ml min⁻¹. Electrochemical measurements of alkaline water electrolysis were conducted by the electrochemical station (CS350M + CS2020B, Wuhan Corrtest Instruments Co., Ltd.; the two devices are integrated together). The polarization curve at several temperatures (50°, 60°, 70°, and 80°C) in the voltage range of 1.3 ~ 2.6 V were recorded. EIS was also monitored by the same electrochemical workstation in the frequency from 10 kHz to 0.1 Hz at different temperatures to evaluate the ohmic resistance. The definition of the current density improvement rate is as follows: Under the same conditions, subtract the current density of the serpentine configuration from that of the array configuration, and then, divide the resulting value by the current density of the serpentine configuration.

In situ observation bubble system

To observe the movement of bubbles between three different flow channels during the alkaline water electrolysis process, a transparent quartz glass plate embedded with silver wires was used to fabricate the water electrolyzer. A high-speed camera (HF Agile Device) and in situ observation system were established. To precisely capture the visual images of bubble generation and release behavior in the flow channels during the water electrolysis process, transparent quartz glass was used to replace the traditional metal cell, enabling clear bubble imaging. Grooves were etched inside the quartz glass to embed silver wires, which were directly connected to the metal catalyst during actual testing to enhance conductivity. Through these device improvements, the bubble escape behavior during the operation of the alkaline electrolyzer can be observed in situ using a high-speed camera (Frames per second at resolution = 10,000, 1280 by 1024). Considering the limited resistance of quartz glass to strong alkalis, a 1 M KOH solution was used in place of the 30 wt % KOH for the in situ bubble observation process.

Scaled-up experiment

The membranes and catalysts used in the scaled-up water electrolysis experiments are the same as the small-sized electrolyzer. The membrane was provided by Shenzhen KeXin Hydrogen Energy Co., Ltd., and the anode used commercial NiFe foam as the catalyst, while NiMo foam was used for the cathode. The electrolyte was also 30 wt % KOH solution with a flow rate of 100 ml min⁻¹. Due to the ultra-high current of 100 A exceeding the upper limit of the electrochemical workstation's range, a dc power supply was used to deliver constant

current, while a voltmeter was used to record the voltage values at the corresponding current levels. The experimental temperature was maintained at 80°C using two programmable heaters, while voltage measurements were recorded at current intervals of 10 A from 10 to 100 A.

Supplementary Materials

The PDF file includes:

Supplementary Methods

Figs. S1 to S56

Tables S1 to S16

Legends for movies S1 to S3

References

Other Supplementary Material for this manuscript includes the following:

Movies S1 to S3

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Machine learning–driven discovery of optimal designs for water electrolysis devices

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